

M3 · LightGBM (gradient boosting)

Model 4 of the ladder · the main candidate · AI Credit Risk / Loan Approval Engine

In one sentence: boosting builds trees **one after another**. Each new small tree is fitted to the mistakes that all previous trees together still make, and is added with a small weight (the learning rate). The sum of all trees is the log-odds of default, which the sigmoid turns into a PD. Random Forest averages independent deep trees; boosting uses many shallow trees that **cooperate**.

How it was set up

- Same feature sets as M1/M2: **A** 189, **B** 147 features (no LendingClub grades; deployable). No scaling needed.
- **Run on Kaggle** (private CPU notebook, 4 cores, LightGBM 4.6.0), driven from the CLI; only train and validation were uploaded, so the test set stays sealed.
- **Random search:** the same 40 settings for A and B at learning rate 0.05 (55 minutes). Each fit stops when validation AUC has not improved for 100 rounds. Validation therefore both stops and selects, so its score is slightly optimistic; the sealed test set will give the honest number.

Chosen parameters

Parameter	Version A	Version B	What it means
num_leaves	20	21	Leaves per tree (tree complexity). Searched 15-255.
min_data_in_leaf	442	1433	Minimum loans per leaf (regularisation). Searched 20-2,000.
feature_fraction	0.569	0.639	Share of features each tree may use. Searched 0.4-1.0.
bagging_fraction	0.83	0.65	Share of loans each tree trains on. Searched 0.5-1.0.
lambda_l1	0.4315	0.02773	L1 penalty on leaf values. Searched 0.0001-10.
lambda_l2	0.697	0.2062	L2 penalty on leaf values. Searched 0.0001-10.
min_gain_to_split	0.054	0.478	Minimum loss improvement to make a split. Searched 0-0.5.
learning_rate / rounds	0.02 / 2961	0.02 / 2279	Step size and the number of trees chosen by early stopping (100-round patience).

Experiments (validation)

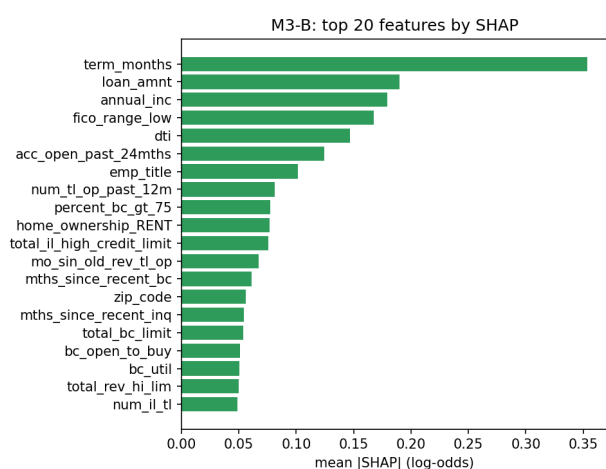
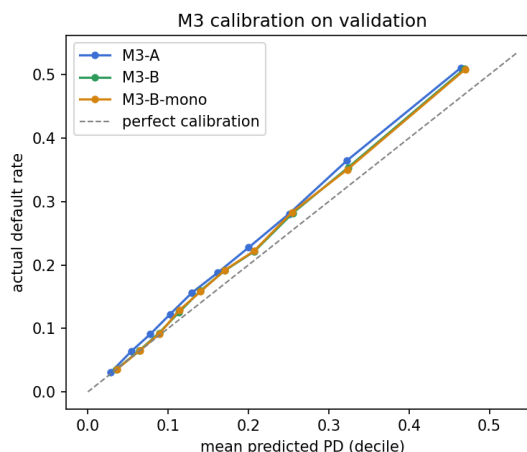
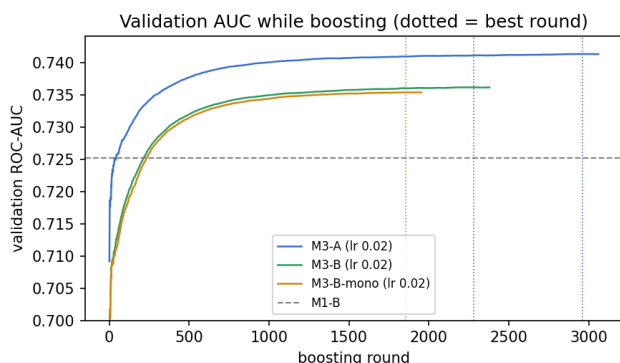
Experiment	Version A	Version B	Takeaway
Untuned baseline → best of search (ROC-AUC)	0.7403 → 0.7410	0.7346 → 0.7356	Tuning adds a little; the model family does the heavy lifting.
Learning rate 0.05 vs 0.02 (ROC-AUC)	0.7406 vs 0.7414	0.7356 vs 0.7362	Kept: 0.02 (A), 0.02 (B); 0.02 only if +0.0005.
is_unbalance=True: ROC-AUC / Brier	0.7411 / 0.2059	0.7362 / 0.2149	Up-weighting defaults ruins probabilities again; not used.
Monotonic constraints (B): ROC-AUC	-	0.7354 (-0.0008)	FICO ↓, DTI ↑, income ↓, 60-month ↑, inquiries ↑, utilisation ↑ risk.

Results (validation unless marked)

Model	ROC-AUC	KS	PR-AUC	Brier	Log loss	Mean PD	Actual
M0	0.5000	0.0000	0.2035	0.1631	0.5085	17.24%	20.35%
M1-A / M2-A	0.7335 / 0.7345	0.3421	0.4120	0.1445	0.4506	17.73%	20.35%
M3-A	0.7414	0.3519	0.4250	0.1422	0.4448	17.92%	20.35%
M1-B / M2-B	0.7252 / 0.7270	0.3262	0.4045	0.1449	0.4529	18.37%	20.35%
M3-B	0.7362	0.3415	0.4210	0.1426	0.4463	18.69%	20.35%
M3-B-mono	0.7354	0.3400	0.4201	0.1428	0.4467	18.70%	20.35%
M3-B (train)	0.7567	0.3773	0.4036	0.1242	0.3980	17.25%	17.24%

M1/M2 rows show ROC-AUC for both; the other M1/M2 columns are M1's. All sanity checks passed (early stopping triggered, beats M0, below the 0.80 leakage flag).

Verdict: yes, boosting wins. LightGBM beats Logistic Regression by **+0.0110** ROC-AUC on the deployable version B (above the +0.005 'clear gain' bar) and Random Forest by +0.0092, with a better Brier score (0.1449 → 0.1426). Non-linear effects do matter once trees correct each other's mistakes. M3 is the leading candidate.



Why one risky applicant got PD 53.5% (average 18.7%; actually defaulted: True)

Feature	Value	SHAP (log-odds)
term_months	60	+0.642
acc_open_past_24mths	12	+0.339
hs		
fico_range_low	660	+0.254
emp_title	0.2329	+0.209
inq_last_6mths	4	+0.166
mths_since_recent_bc	1	+0.145
percent_bc_gt_75	11.1	-0.134

Positive = pushed risk up. These per-applicant contributions are what the decision layer will turn into reason codes.

Key findings

- **Ranking:** version B ROC-AUC 0.7252 (M1) → 0.7270 (M2) → **0.7362** (M3); version A 0.7335 → 0.7414 (+0.0079). Approving the safest 50% gives a bad rate of 9.56% with M3-B vs 10.03% with M1-B (M0: 20.35%).
- **Monotonic constraints cost +0.0008 ROC-AUC** on B. The constrained model can never say "higher FICO = riskier", which makes it easier to defend to a credit committee or regulator; it is a strong option for deployment.
- **Little overfitting:** train ROC-AUC 0.7567 vs validation 0.7362 for B (Random Forest: 0.855 vs 0.727). Early stopping on validation limits how far the trees can memorise the training loans.
- **Calibration:** mean PD 18.69% vs actual 20.35%; the riskiest decile is predicted at 46.9% vs 50.9% actual. Same 2015 drift as before, so calibration on validation is still a required step before deployment.
- **LendingClub's grades:** A beats B by 0.0052 ROC-AUC. **Segments (B):** ROC-AUC 0.701 (36-month) vs 0.686 (60-month). Rank agreement with M1-B: Spearman 0.934.